



ROOT CAUSE UNLOCKED · CONDITION KIT

The Brain Health Root Cause *Kit*

The treatable causes worth ruling out first, the tests that are not blood tests, and the one panel to be skeptical of

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The Brain Health Root Cause Kit

The treatable causes worth ruling out first, the tests that are not blood tests, and the one panel to be skeptical of

BEFORE YOU START

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This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

If you are in crisis right now, if you are thinking about harming yourself or ending your life, put this guide down and call or text **988** (the Suicide and Crisis Lifeline, available 24/7). Nothing in any lab panel matters more than that call.

Same-day medical attention, not a lab plan. Sudden confusion, sudden severe headache, one-sided weakness or facial droop, trouble speaking, a first seizure, new memory loss after a head injury, or hallucinations. These are emergencies.

Nothing in this guide is a reason to stop, skip, or reduce a psychiatric medication. Stopping an antidepressant or any psychiatric prescription abruptly can cause genuine withdrawal effects and worse. Every medication decision belongs with your prescriber.

The nine keys, and what tests each one

Unlock Your Brain Health is organized around nine keys. Here is how each maps to something measurable, and where no blood test exists at all.

Key	What it means	How you find it
1. Signals, not verdicts	Brain fog and low mood are symptoms with causes	The workup below, not a self-diagnosis
2. Treatable causes first	The highest-yield chapter in the book	This entire guide
3. Gut-brain axis	Real science, often oversold	Symptoms pattern; see the honesty section below
4. Blood sugar	The brain runs on stable fuel	Fasting glucose, HbA1c, fasting insulin
5. Inflammation	A quiet driver of mood symptoms	hs-CRP
6. Sleep	The nightly cleaning shift	A sleep study if apnea signs. Not a blood test
7. Stress and the HPA axis	A physical feedback loop	History and pattern. Not a blood test
8. Nutrition and movement	The two strongest levers	The diet guide. Not a blood test
9. Longer-horizon terrain	Infections, toxicants, hearing	Targeted tests where history points

Three of the highest-value checks in this guide are not blood draws at all. They are a sleep study, a hearing screen, and a medication review. They appear below with the labs because they belong in the same workup.

The treatable-causes workup, item by item

Every item here is common, testable, and genuinely correctable. Finding one true item on this list is worth more than months of supplements aimed at a problem that was never the driver.

B12, and the two markers that catch what B12 misses

Vitamin B12 deficiency produces a well-documented range of neuropsychiatric symptoms, including depression, cognitive slowing, and memory difficulty (PMID 35337631). It is common in older adults, in long-term users of acid-reducing medications, in vegans and vegetarians, and after bariatric surgery.

The testing subtlety that matters: serum B12 does not always reflect what your cells are actually getting. Methylmalonic acid (MMA) is the most sensitive functional marker of B12 status, and homocysteine rises when B12, folate, or B6 run short. A B12 in the low-normal range with an elevated MMA is a functional deficiency that a B12-only test would have waved through. One caution from my own interpretive framework: MMA also rises when kidney function is impaired, so it is read alongside creatinine.

Folate, with a U-shaped surprise

Ask for red blood cell folate rather than serum folate; it reflects roughly three months of tissue status rather than this week's meals. And note what a national analysis in U.S. women found: both the lowest and the highest folate levels were associated with higher odds of depression compared with the middle (PMID 41824854). Nutrient status is about sufficiency, not maximizing a number, which is a sentence worth applying to everything in this guide.

Thyroid, with both halves of the truth

Hypothyroid brain fog is a genuine, recognized clinical experience, with fatigue, low mood, and executive-function complaints that often precede diagnosis (PMID 35414261). And the honest other half: an analysis pooling individual data from more than 74,000 adults found subclinical thyroid dysfunction was not associated with measurable cognitive differences at the population level (PMID 34491268). Both can be true. A TSH and free T4 are simple, inexpensive tests; correcting a genuinely abnormal result is worthwhile on its own terms. Thyroid targets are individual enough, by age, sex, and treatment status, that this guide deliberately publishes no thyroid reference table; that conversation belongs with your clinician over your actual numbers.

Blood count and iron

Low hemoglobin means less oxygen delivery to the brain, and the cognitive association is measurable well above dangerously low levels, with a larger effect in older adults (PMID 42504232). A complete blood count plus ferritin catches most of it. Ferritin is also an acute-phase reactant, so it reads alongside hs-CRP: a “normal” ferritin during inflammation can hide depleted iron.

Vitamin D

Low vitamin D has been linked to depressive symptoms, and the honest grade from the review literature is that supplementation trials for mood are inconsistent: worth checking and correcting if low, not guaranteed to fix anything (PMID 38931257).

Blood sugar, three ways

Fasting glucose, HbA1c, and fasting insulin together read the fuel-stability story from Key 4. The brain notices insulin resistance years before a diabetes diagnosis, and brain fog that is worse after meals or better with regular eating is a pattern worth bringing up specifically.

Inflammation

hs-CRP is the inexpensive rough read on systemic inflammation, which has documented links to mood symptoms in a meaningful subset of people (PMID 26711676). A single elevated reading after illness means little; repeat it when well.

Homocysteine, read correctly

Homocysteine is a methylation-cycle marker. It rises when B12, folate, or B6 run short, and with thyroid or kidney trouble. Do not let anyone wave off an elevated result as trivia. In my read this is one of the most consequential numbers in the panel, because it tells you the methylation machinery your brain chemistry depends on is running short somewhere findable. The way to act on it is not to chase the number with blind supplementation. Find which input is short, B12, folate, B6, thyroid, or kidney function, correct that, and let homocysteine confirm the fix on retest. My functional target of 5 to 7 in the table below reflects how seriously I take it. Note that it is a window, not a floor. Lower is not better: a homocysteine driven below that window points toward overmethylation and B vitamin intolerance, which is its own problem rather than a better result.

The three checks that are not blood tests

A sleep study, if the signs are there. Loud snoring, witnessed pauses, gasping awake, or unrefreshing sleep deserve a real sleep evaluation. Self-reported sleep apnea was associated with poorer memory and higher dementia-risk scores in cognitively healthy middle-aged adults (PMID 42370433), and apnea is directly treatable, which makes it one of the highest-value items in this entire guide.

A hearing screen, especially past 50. The Lancet Commission identified midlife hearing loss as one of the most significant modifiable dementia risk factors across the life course (PMID 32738937), and audiometry-based U.S. data attributed an estimated 17 percent of dementia cases among those with moderate or greater hearing loss to the hearing loss itself (PMID 37139951). A hearing screen is quick, painless, and almost never on anyone's brain fog list. Put it on yours.

A medication review, including over-the-counter. Anticholinergic burden, the cumulative load of common allergy, bladder, sleep, and some antidepressant medications, is associated with measurably more cognitive impairment at higher burden (PMID 42156261). Bring every bottle, including supplements, to one appointment and ask the specific question: "What is my total anticholinergic burden?"

The panel to be skeptical of: urinary neurotransmitter testing

If you have been offered a urinary neurotransmitter panel as a readout of your brain chemistry, here is the honest picture I owe you. The chemicals are real, and the problem lies in the interpretation rather than the lab work. Most serotonin is made in the gut, serotonin does not meaningfully cross the blood-brain barrier, and urinary dopamine, norepinephrine, and epinephrine come overwhelmingly from the adrenal glands, gut, and

peripheral nerves, not from inside your skull. The scientific review that examined this question directly concluded the established clinical use of such testing is detecting a rare adrenal tumor, and that the evidence connecting urinary levels to mood or attention was early and limited (PMID 20696183). A number on a urine panel is one data point collected outside your brain. It should never be sold to you as a brain-chemistry readout or serve as the sole basis for a diagnosis or treatment plan, here or anywhere.

Functional ranges for the markers in this guide

Clinical targets I use at Root Health, not validated diagnostic thresholds, narrower than the standard laboratory range on purpose. Thyroid values are deliberately absent (see above); hemoglobin ranges differ by sex, so the row below is approximate and your lab's sex-specific range governs.

Marker	Units	Functional target	Approximate standard range
Vitamin B12	pg/mL	500 to 900	200 to 1100
Methylmalonic acid (MMA)	nmol/L	under 162	under 243
Folate (RBC)	ng/mL	600 to 1500	400 to 2000
Homocysteine	umol/L	5 to 7	under 10 to 15, varies by lab
Vitamin D (25-OH)	ng/mL	40 to 80	20 to 100
Hemoglobin	g/dL	13.0 to 17.5	10 to 18, sex-specific
Ferritin	ng/mL	50 to 150	25 to 300
Glucose, fasting	mg/dL	75 to 86	70 to 99
HbA1c	%	4.6 to 5.3	4.0 to 5.6
hs-CRP	mg/L	under 0.5	under 1.0

Reading your results as a pattern

Pattern one: a treatable cause found. Low B12 with high MMA, an abnormal thyroid, anemia, sleep apnea, a heavy anticholinergic list. The best outcome this guide can produce is a specific, correctable driver, addressed with your clinician, before any supplement conversation.

Pattern two: everything normal, symptoms real. Common, and not a dead end. It moves the attention to the keys that have no blood test, which are sleep, stress, movement, nutrition pattern, hearing, and the gut-brain terrain. That is where the diet guide and the daily work live.

Pattern three: the metabolic cluster. Creeping glucose, HbA1c drifting, low vitamin D, rising hs-CRP. The brain complaint and the metabolic picture are usually one story; Key 4's terrain work serves both.

Pattern four: mood symptoms dominating. Labs support the workup; they do not replace mental health care. Therapy and, where prescribed, medication are first-class treatments, and everything in this kit is designed to work alongside them, never instead of them.

The conversation with your doctor

A request list you can bring

- B12 with **methylmalonic acid**, and red blood cell folate
- TSH and free T4
- Complete blood count and ferritin
- Vitamin D (25-OH)
- Fasting glucose, HbA1c, fasting insulin
- hs-CRP
- Homocysteine, read as a B-vitamin, thyroid, and kidney workup trigger
- A medication review for anticholinergic burden, with every bottle on the table
- A sleep study referral if snoring, gasping, or unrefreshing sleep applies
- A hearing screen, especially if over 50 or noise-exposed

Your next step

The companion guides do the rest: **The Brain Health Diet Guide** covers the two levers with the strongest evidence in this territory, and **The Brain Health Supplement Guide** covers what genuinely helps, what failed its biggest trial, and the interaction risks that matter most alongside psychiatric medication.

Book a free Root Discovery Session: rootcauseunlocked.com/discovery

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References

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If you are in crisis right now, call or text **988** (Suicide and Crisis Lifeline, 24/7). Food can wait; that call cannot.

Nothing here is a reason to stop, skip, or reduce a psychiatric medication. The strongest diet trial in this territory tested nutrition ADDED to standard care, never instead of it, and that is exactly how this guide is meant to be used.

This guide makes no dementia prevention or reversal claims. Where the research is observational or modest, it says so.

The one idea underneath everything

Key 8 of *Unlock Your Brain Health* says it plainly: the two interventions with the deepest evidence for brain fog and mood are the least exciting ones, what you eat on a normal Tuesday and whether you move regularly. Supplements layered on a poor foundation are the most common and expensive mistake in this territory. This guide is the foundation half, and it is deliberately boring in the way the evidence is boring.

Part one: the two trials worth knowing by name

MIND, for the aging brain. In a cohort of older adults followed for years, those whose eating most resembled the MIND pattern, a hybrid of Mediterranean and DASH emphasizing leafy greens, other vegetables, berries, nuts, whole grains, fish, and olive oil, declined cognitively at a substantially slower rate, a difference the researchers described as equivalent to being 7.5 years younger (PMID 26086182). Honest grade. Observational, an association rather than proof, and among the strongest and most consistent associations in this entire field.

SMILES, for mood. A randomized controlled trial tested a dietitian-supported Mediterranean-style dietary improvement program added to participants' existing depression care against a social-support control. The diet group improved significantly more, with roughly one in three reaching remission versus one in twelve in the control group (PMID 28137247). Honest grade. A real randomized trial, modest size, tested as an addition to standard care. That is the model: food as a treatment partner, not a replacement.

What the two trials agree on is almost the same plate: vegetables at most meals with leafy greens most days, berries regularly, fish and legumes as protein anchors, nuts and olive oil as default fats, whole grains over refined, and a sharp drop in ultra-processed food. Nobody's brain plate needs to be more exotic than that.

Part two: fuel stability, the every-three-hours story

Key 4's blood sugar biology translates to the plate simply. The brain runs poorly on fuel spikes and crashes, and the classic pattern, foggy and irritable late morning after a carbohydrate breakfast, better after regular balanced meals, is worth testing on yourself for two weeks:

1. **Protein at every meal, breakfast especially.** Eggs, fish, yogurt, legumes, meat: any of them beats toast alone.
2. **Never carbohydrate naked.** Pair it with protein, fat, or fiber to flatten the curve.
3. **Regular meal timing** while symptoms are active; skipped meals are a common, cheap-to-fix fog trigger.
4. **Front-load the day.** A substantial breakfast and lunch with a lighter dinner supports both fuel stability and sleep.

If the lab guide's glucose, HbA1c, or fasting insulin numbers were drifting, this part is doing double duty for brain and metabolism at once.

Part three: the specific foods with brain-specific reasons

Fish, especially the fatty ones. Salmon, sardines, trout, and herring supply EPA and DHA. The supplement guide covers the trial evidence for EPA in diagnosed depression; the food-first version is two fish meals a week, which is also the MIND pattern's floor.

Leafy greens most days and berries most weeks: the two food groups the MIND pattern weights hardest, and the easiest to systematize (a standing salad, frozen berries on breakfast).

Fermented and fiber-rich foods for the gut-brain terrain. Key 3's honest framing applies: the gut-brain axis is genuine science that is still maturing, and the defensible food translation is unspectacular. Fiber variety feeds the microbes that make short-chain fatty acids, and yogurt, kefir, and fermented vegetables are reasonable inclusions rather than miracle foods.

Coffee and tea, timed. Caffeine is fine for most brains and genuinely hostile to sleep when late; a noon cutoff is the cheap experiment for anyone with both fog and poor sleep.

Alcohol, honestly. It fragments sleep architecture even when it speeds falling asleep, and poor sleep is a direct fog generator (the ebook's Key 6 covers the glymphatic cleaning story). If evenings include alcohol and mornings include fog, that pairing deserves a two-week trial separation before any supplement does.

Hydration, plainly. Even mild dehydration reads as fatigue and poor concentration for many people; pale-straw urine is the practical gauge.

What this guide deliberately does not include

This guide leaves out “brain detox” diets, ketogenic-cures-Alzheimer's claims (the ebook's Key 4 explains why the “type 3 diabetes” nickname is shorthand for real research, not a diagnosis or a protocol), elimination diets without a specific tested reason, and any food list that promises to raise a neurotransmitter. The two trials in part one are the honest ceiling of what nutrition evidence currently supports here, and they are plenty.

What a day looks like

- **Breakfast.** Eggs with sauteed greens; oatmeal with frozen berries and walnuts on the side.
- **Lunch.** A big salad with salmon or lentils, olive oil, whatever vegetables exist.
- **Dinner.** Any protein, two vegetables, beans or a whole grain, lighter than lunch.
- **Rhythm.** Fish twice a week, greens most days, berries most weeks, caffeine before noon, alcohol watched honestly, water to pale straw.

Working with Root Health

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Read the companion documents alongside this one. **Understanding Your Brain Fog and Mood Labs** covers the treatable causes to rule out first, and **The Brain Health Supplement Guide** covers what genuinely helps beyond the plate.

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1. Morris MC, Tangney CC, Wang Y, et al. MIND diet slows cognitive decline with aging. *Alzheimers Dement*. 2015;11(9):1015-1022. PMID 26086182
2. Jacka FN, O'Neil A, Opie R, et al. A randomised controlled trial of dietary improvement for adults with major depression (the 'SMILES' trial). *BMC Med*. 2017;15(1):23. PMID 28137247

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If you are in crisis right now, call or text **988** (Suicide and Crisis Lifeline, 24/7). No supplement has any role in a crisis.

Nothing in this guide is a reason to stop, skip, or reduce a psychiatric medication. Every entry below is written as a possible addition to standard care, coordinated with your prescriber, never as an alternative to it. Stopping psychiatric medication abruptly can cause genuine withdrawal effects and worse.

The interaction section below is the most important part of this guide. If you take an antidepressant of any kind, read it before anything else.

This guide makes no dementia prevention or reversal claims, and it names the trial that taught everyone why.

Supplements in the United States are regulated as food, not as drugs. Quality varies. Look for independent third-party testing.

The serotonin interactions that outrank everything else here

Two popular mood supplements have real pharmacology, which is exactly why they carry real risk alongside psychiatric medication.

St. John's Wort has documented, clinically significant interactions with SSRIs and other antidepressants, with oral contraceptives, with blood thinners, and with a long list of prescriptions, including the risk of serotonin syndrome when combined with serotonergic antidepressants (PMID 41901246). It also reduces the effectiveness of several medications by speeding their breakdown, which is a quiet way to lose contraception or anticoagulation without noticing.

5-HTP raises serotonin directly, and combining it with an SSRI, SNRI, or MAOI carries a real risk of serotonin syndrome, with agitation, rapid heart rate, tremor, and in severe cases a medical emergency (PMID 30599219). **SAMe** sits in the same caution category alongside antidepressants.

The rule, with no exceptions this guide will endorse. None of these three starts, continues, or combines with a psychiatric prescription without your prescriber's explicit sign-off, and even without a prescription in the picture, they belong in an ask-first conversation, not a cart. If you develop agitation, racing heart, tremor, sweating, or confusion on any serotonin-active combination, that is an emergency, not a side effect to push through.

Why nutrient evidence looks thinner than it is

A large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back. You cannot patent fish oil or a B vitamin, so the evidence hierarchy over-documents patentable drugs and under-documents nearly everything else. A Cochrane methodology review found manufacturer sponsorship produces more favourable results and conclusions than independent funding, through a bias standard quality checks do not catch (PMID 28207928). So when this guide says “no evidence,” it will say whether it means *tested and failed*, *never properly tested*, or *tested small and won*. The asymmetry earns nutrients humility, not automatic credit: small nutrient trials carry their own publication bias toward positive findings. One rule, both directions. This category also has a unique stake in the rule, because the brain supplement aisle is the industry’s most profitable fog.

Tier 1: what actually has human evidence

EPA-predominant omega-3, for diagnosed depression, alongside care

The omega-3 literature looks contradictory until one distinction is applied, and then it organizes. A meta-analysis of 35 randomized trials found that in people with diagnosed depression, formulations that were predominantly EPA (more than half EPA) showed clinically meaningful benefit over placebo, while DHA-predominant formulations did not, and omega-3 did not prevent depressive symptoms in people without depression (PMID 27103682).

Three honest boundaries follow directly from that finding. This is evidence for **EPA-predominant products** specifically, so read the label ratio rather than the word “fish oil.” It is evidence in **diagnosed depression as an addition to care**, coordinated with whoever manages that care. And it is **not** evidence for prevention or for fog without mood symptoms, where the food-first version (two fish meals weekly, per the diet guide) remains the defensible play. Practical note: omega-3 at supplement doses has a blood-thinning effect worth naming to your prescriber if you take anticoagulants or face surgery.

Correcting measured deficiencies in B12, folate, and vitamin D

This tier's least glamorous entry is its most defensible: when the lab guide's workup finds a genuine deficiency, correcting it is evidence-based medicine, not supplementation culture. B12 deficiency produces neuropsychiatric symptoms that respond to repletion (PMID 35337631); folate status relates to depression risk in a U-shape that argues for sufficiency, not megadosing (PMID 41824854); and vitamin D is worth checking and correcting if low, with the review literature candid that supplementation trials for mood in non-deficient people are inconsistent (PMID 38931257). Forms matter modestly (methylated B vitamins for those who need them; vitamin D with food), doses follow the actual lab result, and retesting closes the loop. What this tier is not: a reason to take any of these at high doses with normal labs.

Tier 2: tested and failed, named honestly

Ginkgo biloba, for the aging brain

Ginkgo is the best-selling memory herb in the world, and it received exactly the trial its popularity deserved. The GEM trial randomized 3,069 older adults against placebo and followed them for a median of six years. Ginkgo did not reduce the incidence of dementia or Alzheimer's disease (PMID 19017911). That is what *tested and failed* looks like at full scale, on public money (the trial was funded by the NIH). If ginkgo is in your cabinet for memory protection, that is the study to weigh it against.

The nootropic blend shelf

Multi-ingredient “brain support,” “focus,” and “nootropic” stacks are the supplement industry’s fastest-growing fog. Their individual ingredients range from *never properly tested* to *tested small*, their combinations are essentially never tested at all, and blend labels routinely underdose the one ingredient with any evidence behind it. The same rule applies as in the rest of this product line. Every ingredient is chosen on purpose, at a dose you know, for a reason you can say out loud. A blend fails that rule by construction.

What actually protects the brain is not in a capsule

The honest close of this guide points back out of it. The strongest interventions in this territory are the diet pattern and the movement habit (the diet guide carries their trials), sleep treated as the non-negotiable it is, hearing checked and corrected, blood pressure and blood sugar managed, and the treatable causes in the lab guide actually ruled out. The Lancet Commission’s modifiable dementia risk factors are lifestyle and medical factors, not supplements (PMID 32738937). A reader who does those things and takes nothing from this guide’s shelves is better protected than one who inverts the order.

What I deliberately left out

Racetams and other gray-market “smart drugs,” high-dose single amino acids for neurotransmitter loading, “brain detox” products, and every blend whose mechanism is described in marketing language rather than chemistry. A guide reaches you without a history, a medication list, an exam, or any way for me to change course next week, so the bar for what goes in it is higher than the bar for a conversation in my office. None of those clear it.

Where to find quality versions of these

Quality varies; look for independent third-party testing (NSF, USP, or equivalent) whichever source you use. I maintain a professional dispensary at Root Health through Fullscript:

us.fullscript.com/welcome/drseay

Disclosure: Root Health earns a portion of dispensary purchases. Nothing here was included, excluded, ranked, or dosed based on what it earns. Notice that the most powerful tools in this guide, sleep, movement, the diet pattern, a hearing check, and the 988 line when it is needed, cost nothing and earn me nothing. That is where I want you to start anyway.

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